Lab 1: Requirements Engineering & UML Use Case Modelling

Student	Aniyath Amrita Pradeep	**SRN**	PES2UG24CS919
Problem Statement	Community Lost & Found Matching Network	**Domain**	Media, Events & Community

1. Problem Context and Overview

The Community Lost and Found Matching Network is a platform that lets residents of a housing community, campus, or neighbourhood report items they have lost or found. The system compares new reports against existing listings using category, location, and descriptive keywords, and surfaces likely matches to the people involved. A finder and an owner can exchange enough information through the platform to arrange a handover, while a community admin keeps the listings accurate and removes reports that look fraudulent or abusive.

Target Stakeholders / Actors: **Finder / Owner** (a community member who reports a lost or found item, browses listings, and claims a match), **Community Admin** (a moderator who reviews flagged listings and manages community-level reports).

2. Functional Requirements

ID	Requirement	Priority	Acceptance Criteria	Rationale
FR-001	The system shall allow a Finder or Owner to submit a lost or found item report with category, location, date, description, and at least one photo.	High	Pass: report saved with a unique listing ID and appears in search within 5 seconds. Fail: report saved without a mandatory field or without an associated photo.	A complete report is the only reliable basis the matching logic has to work from.
FR-002	The system shall generate ranked match suggestions for a new report by comparing category, location radius, and description keywords against the opposite report type.	High	Pass: a found item within 2 km and same category appears in the owner's match list ranked by similarity score. Fail: an exact category and location match is not surfaced.	Automated matching is the core value the platform adds over a plain classifieds board.
FR-003	The system shall let a Finder or Owner initiate a claim on a suggested match and route it through an identity verification step before contact details are shared.	High	Pass: claim moves to Pending Verification and contact details stay hidden until verification passes. Fail: contact details are shared before verification completes.	Unverified handovers are the main way lost and found platforms get exploited.
FR-004	The system shall allow a Community Admin to review, edit the status of, or remove any listing that has been flagged by a member or by automated abuse checks.	Medium	Pass: a flagged listing is hidden from search once the admin marks it removed, and the action is recorded with a timestamp and admin ID. Fail: a removed listing still appears in search results.	Community trust depends on visible moderation, particularly once claims involve real handovers between strangers.
FR-005	The system shall notify a Finder or Owner by email or in-app alert whenever a new match is found for their active listing or when the status of an existing claim changes.	Medium	Pass: a notification is dispatched within 2 minutes of a qualifying match or status change. Fail: a status change occurs with no corresponding notification sent.	Item recovery is time sensitive, so users need to learn about a match as soon as it exists.

3. Non-Functional Requirements

ID	Type	Requirement	Priority	Acceptance Criteria	Rationale
NFR-001	Performance	Match suggestion queries shall return ranked results across up to 500,000 historical listings in under 2 seconds at the 95th percentile.	High	Pass: benchmark run against a 500,000 row seeded dataset returns results in under 2 seconds for 95 percent of queries. Fail: p95 latency exceeds 2 seconds under simulated load.	Slow matching defeats the purpose of the platform, since users expect near real time results when checking for their item.
NFR-002	Security & Privacy	The system shall not expose a claimant's exact address or phone number to the other party until identity verification for that claim is marked complete, and all stored contact details shall be encrypted at rest.	High	Pass: API responses for an unverified claim omit precise address and phone fields, and a database dump shows encrypted contact fields. Fail: raw contact details are visible before verification or found unencrypted in storage.	The platform connects strangers around physical handovers, so premature exposure of personal contact details is a direct safety risk.

4. UML Use Case Diagram

The diagram below models both actors, the primary use cases each actor initiates, and the internal use cases they depend on. Report Lost Item and Report Found Item each include Upload Item Media, since a report cannot be submitted without at least one photo. Claim Item includes Verify Claimant Identity, reflecting FR-003. View Match Suggestions includes Send Match Notification, reflecting FR-002 and FR-005. Flag Suspicious Listing extends Browse and Search Listings as an optional path available only when a member chooses to report a listing while browsing.

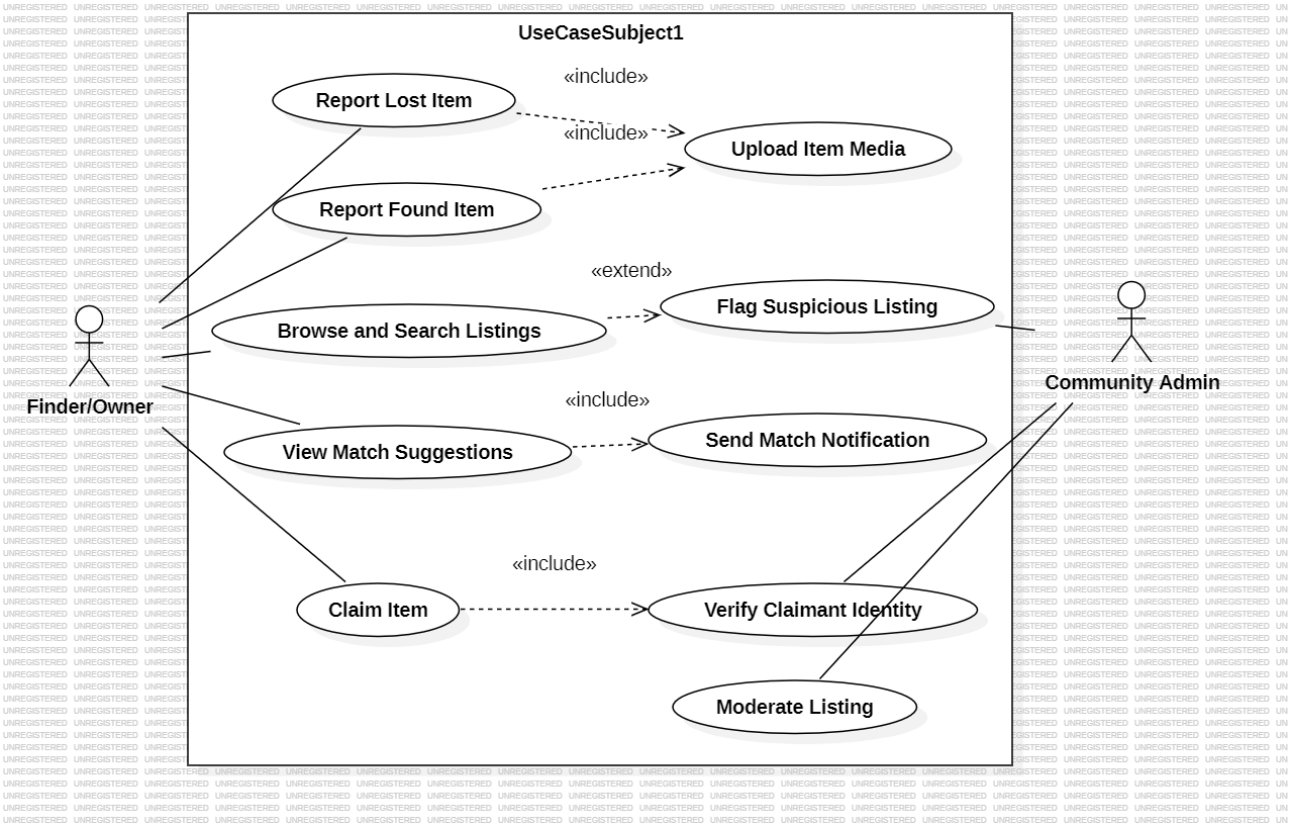


Figure 1. Use case diagram for the Community Lost and Found Matching Network.

5. Use Case Flow Specification

Use Case: Claim Item

Actors. Finder / Owner (primary), Community Admin (secondary, notified on disputed claims)

Trigger. The Finder or Owner selects a suggested match from their listing and chooses to initiate a claim.

Preconditions

The user has an active account and at least one active listing (lost or found report). A match with a similarity score above the system threshold exists for that listing and has not already been claimed by another user.

Postconditions

On success, the claim record moves to Verified status, both parties receive each other's contact details, and the two listings are marked as Matched so they no longer appear in active search results.

Main Success Scenario

1. The user opens View Match Suggestions and selects a listing from the ranked results.
2. The user chooses Claim Item on the selected match.
3. The system executes the included use case Verify Claimant Identity, prompting the user for a short description of a unique, non-obvious detail about the item.
4. The system compares the submitted detail against the private fields recorded on the original listing and marks verification as passed.
5. The system creates a claim record linking both listings and sets its status to Verified.
6. The system executes the included use case Send Match Notification, alerting both parties that verification succeeded and sharing each party's contact details.
7. Both listings are marked Matched and removed from active search results.

Alternate Flow. Verification Fails

1. At step 4 of the main scenario, the submitted detail does not match the private fields on the original listing.
2. The system sets the claim status to Verification Failed and allows the user up to two further attempts.
3. If a third attempt also fails, the system locks further claims on that match for 24 hours and flags the claim for Community Admin review.
4. The Community Admin may manually approve or reject the claim after checking the reports outside the app, for example through the listing's photos and description history.
5. The use case resumes at step 5 of the main scenario if the admin approves the claim, or ends without a matchif the admin rejects it.